

2520030340

OS PRACTICAL – 7

PART A — MEMORY ADDRESS SPACE ANALYSIS

AIM

To write a program that prints the addresses of **code, global, static, heap, and stack variables**, analyze the Linux process address space layout, and explain the purpose of each memory segment.

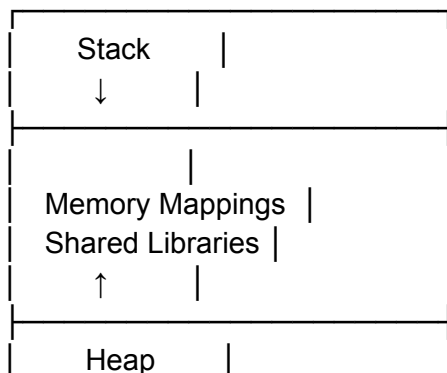
OBJECTIVES

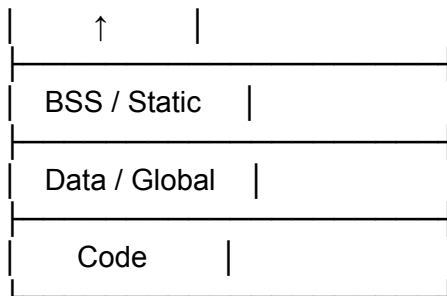
1. To understand the memory layout of a Linux process.
 2. To print the address of a code/function variable.
 3. To print the address of global and static variables.
 4. To print the address of a heap variable.
 5. To print the address of a stack variable.
 6. To analyze the `/proc/<PID>/maps` file.
 7. To understand virtual memory mapping.
-

MEMORY SEGMENTS

A Linux process generally has memory areas like:

High Address





Low Address

Simple meaning

Segment	Contains
Code/Text	Program instructions
Global/Data	Initialized global variables
BSS	Uninitialized global/static variables
Static	Static variables
Heap	Dynamically allocated memory
Stack	Local variables and function calls

PROGRAM

File name

memory_layout.c

COPY-PASTE CODE

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 100;
static int static_var = 200;

void show_code_address()
{
    printf("Code address      : %p\n", (void *)show_code_address);
}
```

```

}

int main()
{
    int stack_var = 300;

    int *heap_var = (int *)malloc(sizeof(int));

    if (heap_var == NULL)
    {
        perror("malloc");
        return 1;
    }

    *heap_var = 400;

    printf("==== MEMORY ADDRESS SPACE =====\n");

    printf("Process ID      : %d\n", getpid());

    show_code_address();

    printf("Global variable   : %p\n", (void *)&global_var);

    printf("Static variable   : %p\n", (void *)&static_var);

    printf("Heap variable     : %p\n", (void *)heap_var);

    printf("Stack variable    : %p\n", (void *)&stack_var);

    printf("\n==== VARIABLE VALUES =====\n");

    printf("Global value      : %d\n", global_var);

    printf("Static value      : %d\n", static_var);

    printf("Heap value        : %d\n", *heap_var);

    printf("Stack value       : %d\n", stack_var);

    printf("\nProcess is running...\n");
    printf("Press Enter to exit.\n");

    getchar();

```

```
free(heap_var);

return 0;
}
```

PART B — `/proc/<PID>/maps` ANALYSIS

Linux provides information about a running process through the `/proc` filesystem.

For a process with PID `1234`, we can view its memory mappings using:

```
cat /proc/1234/maps
```

This shows the virtual memory regions used by the process.

ALGORITHM

Program

1. Start the program.
 2. Create global and static variables.
 3. Create a local stack variable.
 4. Dynamically allocate memory using `malloc()`.
 5. Print the process ID.
 6. Print the address of the code.
 7. Print the addresses of global and static variables.
 8. Print the heap address.
 9. Print the stack address.
 10. Keep the process running.
 11. Analyze its memory map using `/proc/<PID>/maps`.
 12. Release heap memory.
 13. Terminate the program.
-

NOW — UBUNTU STEP BY STEP

STEP 1 — Create the folder

Open Ubuntu.

Type:

```
mkdir os_practical7
```

Then:

```
cd os_practical7
```

You should see:

```
kruthi@LAPTOP:~/os_practical7$
```

STEP 2 — Create the C file

Type:

```
nano memory_layout.c
```

STEP 3 — Paste the code

Paste the complete `memory_layout.c` code given above.

STEP 4 — Save

Press:

Ctrl + O

Then:

Enter

Then:

Ctrl + X

STEP 5 — Compile

Type:

```
gcc memory_layout.c -o memory_layout
```

If there is **no error**, compilation is successful. 

STEP 6 — Run

Type:

```
./memory_layout
```

You should get something similar to:

```
===== MEMORY ADDRESS SPACE =====
```

```
Process ID      : 700
```

```
Code address    : 0x55xxxxxxxxxx
```

```
Global variable : 0x55xxxxxxxxxx
```

```
Static variable : 0x55xxxxxxxxxx
```

```
Heap variable   : 0x55xxxxxxxxxx
```

```
Stack variable  : 0x7fffffffxxx
```

```
===== VARIABLE VALUES =====
```

```
Global value    : 100
```

```
Static value    : 200
```

```
Heap value      : 400
```

```
Stack value     : 300
```

```
Process is running...
```

```
Press Enter to exit.
```

 **Your addresses will be different. That's completely normal.**



IMPORTANT

When you see:

Process ID : 700

DO NOT PRESS ENTER YET.

The program must remain running because we need its PID to inspect:

/proc/<PID>/maps

STEP 7 — TAKE SCREENSHOT



Take a screenshot showing:

===== MEMORY ADDRESS SPACE =====

Process ID : XXXXX

Code address : ...

Global variable : ...

Static variable : ...

Heap variable : ...

Stack variable : ...

This is your **Program Output Screenshot**.

STEP 8 — OPEN SECOND TERMINAL

Keep the first terminal running.

Open a **second Ubuntu terminal**.

Go to the folder:

cd ~/os_practical7

STEP 9 — View `/proc/<PID>/maps`

Suppose your PID was:

700

Then type:

```
cat /proc/700/maps
```

⚠ Replace 700 with **your actual PID**.

You should see something similar to:

```
55a1c0000000-55a1c0001000 r--p ... /home/kruthi/os_practical7/memory_layout
55a1c0001000-55a1c0002000 r-xp ... /home/kruthi/os_practical7/memory_layout
55a1c0002000-55a1c0003000 r--p ... /home/kruthi/os_practical7/memory_layout
55a1c0003000-55a1c0004000 rw-p ... /home/kruthi/os_practical7/memory_layout
55a1d0000000-55a1d0021000 rw-p ... [heap]
7ffdxxxxxxx-7ffdxxxxxxx rw-p ... [stack]
```

STEP 10 — Easier `/proc/maps` command

Instead of showing everything, you can use:

```
cat /proc/700/maps | grep -E "heap|stack|memory_layout"
```

Again, replace 700 with your PID.

This makes the important regions easier to identify.

You should see entries containing:

```
memory_layout
[heap]
[stack]
```

STEP 11 — TAKE SCREENSHOT

 Take a screenshot of the `/proc/<PID>/maps` output.

This is your:

Memory Mapping Analysis Screenshot

STEP 12 — Check memory information

You can also run:

```
pmap 700
```

Replace `700` with your PID.

If `pmap` is available, it will show the process memory map in a more readable format.

You can also use:

```
pmap -x 700
```

STEP 13 — FINISH THE PROGRAM

Go back to the **first terminal**.

You should still see:

Press Enter to exit.

Press:

Enter

The program terminates.

WHAT TO WRITE FOR ANALYSIS

1. Code Segment

The **code segment** contains the machine instructions of the program.

The address printed using:

```
&show_code_address
```

belongs to the program's code area.

2. Global Segment

The global variable:

```
int global_var = 100;
```

is stored in the program's global/data area.

3. Static Segment

The static variable:

```
static int static_var = 200;
```

has static storage duration and is stored in the program's data/BSS-related area depending on initialization.

4. Heap

The heap is used for dynamically allocated memory.

We allocated memory using:

```
malloc()
```

The address is stored in:

```
heap_var
```

The corresponding region can be seen in:

[heap]

inside `/proc/<PID>/maps`.

5. Stack

The local variable:

```
int stack_var = 300;
```

is stored on the stack.

The stack region appears as:

[stack]

in `/proc/<PID>/maps`.

RESULT

Thus, the addresses of the **code, global, static, heap, and stack variables** were successfully obtained. The Linux process address space was analyzed using the `/proc/<PID>/maps` file and memory analysis tools, and the virtual memory organization of the process was studied successfully.

```
kruthi@LAPTOP-LP9QU3RR: ~$ mkdir os_practical7
kruthi@LAPTOP-LP9QU3RR: ~$ cd os_practical7
kruthi@LAPTOP-LP9QU3RR: ~/os_practical7$ nano memory_layout.c
kruthi@LAPTOP-LP9QU3RR: ~/os_practical7$ ls
memory_layout.c
kruthi@LAPTOP-LP9QU3RR: ~/os_practical7$ gcc memory_layout.c -o m
memory_layout
kruthi@LAPTOP-LP9QU3RR: ~/os_practical7$ ls
memory_layout  memory_layout.c
kruthi@LAPTOP-LP9QU3RR: ~/os_practical7$ ./memory_layout
===== MEMORY ADDRESS SPACE =====
Process ID      : 732
Code address    : 0x5f7a037a2229
Global variable : 0x5f7a037a5010
Static variable : 0x5f7a037a5014
Heap variable   : 0x5f7a04b4f2a0
Stack variable  : 0x7ffc5b334fcc

===== VARIABLE VALUES =====
Global value    : 100
Static value    : 200
Heap value      : 400
Stack value     : 300

Process is running...
Press Enter to exit.
|
```

```

kruthi@LAPTOP-LP9QU3RR: ~
73e21d240000-73e21d243000 rw-p 00000000 00:00 0
73e21d249000-73e21d24b000 rw-p 00000000 00:00 0
73e21d24b000-73e21d24f000 r--p 00000000 00:00 0
[vvar]
73e21d24f000-73e21d251000 r--p 00000000 00:00 0
[vvar_vclock]
73e21d251000-73e21d253000 r-xp 00000000 00:00 0
[vdso]
73e21d253000-73e21d254000 r--p 00000000 08:30 18224
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
73e21d254000-73e21d27f000 r-xp 00001000 08:30 18224
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
73e21d27f000-73e21d289000 r--p 0002c000 08:30 18224
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
73e21d289000-73e21d28b000 r--p 00036000 08:30 18224
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
73e21d28b000-73e21d28d000 rw-p 00038000 08:30 18224
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ffc5b316000-7ffc5b337000 rw-p 00000000 00:00 0
[stack]
kruthi@LAPTOP-LP9QU3RR:~/os_practical7$ cat /proc/732/maps | gre
p -E "heap|stack|memory_layout"
5f7a037a1000-5f7a037a2000 r--p 00000000 08:30 59825
/home/kruthi/os_practical7/memory_layout
5f7a037a2000-5f7a037a3000 r-xp 00001000 08:30 59825
/home/kruthi/os_practical7/memory_layout
5f7a037a3000-5f7a037a4000 r--p 00002000 08:30 59825
/home/kruthi/os_practical7/memory_layout
5f7a037a4000-5f7a037a5000 r--p 00002000 08:30 59825
/home/kruthi/os_practical7/memory_layout
5f7a037a5000-5f7a037a6000 rw-p 00003000 08:30 59825
/home/kruthi/os_practical7/memory_layout
5f7a04b4f000-5f7a04b70000 rw-p 00000000 00:00 0
[heap]
7ffc5b316000-7ffc5b337000 rw-p 00000000 00:00 0
[stack]
kruthi@LAPTOP-LP9QU3RR:~/os_practical7$ |

```



kruthi@LAPTOP-LP9QU3RR: ~



[heap]

7ffc5b316000-7ffc5b337000 rw-p 00000000 00:00 0

[stack]

kruthi@LAPTOP-LP9QU3RR:~/os_practical7\$ pmap -x 732

732: ./memory_layout

Address	Kbytes	RSS	Dirty	Mode	Mapping
00005f7a037a1000	4	4	0	r----	memory_layout
00005f7a037a2000	4	4	0	r-x--	memory_layout
00005f7a037a3000	4	4	0	r----	memory_layout
00005f7a037a4000	4	4	4	r----	memory_layout
00005f7a037a5000	4	4	4	rw---	memory_layout
00005f7a04b4f000	132	4	4	rw---	[anon]
000073e21d000000	160	160	0	r----	libc.so.6
000073e21d028000	1572	992	0	r-x--	libc.so.6
000073e21d1b1000	316	128	0	r----	libc.so.6
000073e21d200000	16	16	16	r----	libc.so.6
000073e21d204000	8	8	8	rw---	libc.so.6
000073e21d206000	52	20	20	rw---	[anon]
000073e21d240000	12	8	8	rw---	[anon]
000073e21d249000	8	4	4	rw---	[anon]
000073e21d24b000	16	0	0	r----	[anon]
000073e21d24f000	8	0	0	r----	[anon]
000073e21d251000	8	4	0	r-x--	[anon]
000073e21d253000	4	4	0	r----	ld-linux-x86-64.s
o.2					
000073e21d254000	172	172	0	r-x--	ld-linux-x86-64.s
o.2					
000073e21d27f000	40	40	0	r----	ld-linux-x86-64.s
o.2					
000073e21d289000	8	8	8	r----	ld-linux-x86-64.s
o.2					
000073e21d28b000	8	8	8	rw---	ld-linux-x86-64.s
o.2					
00007ffc5b316000	132	16	16	rw---	[stack]

total kB 2692 1612 100

kruthi@LAPTOP-LP9QU3RR:~/os_practical7\$ |

```
kruthi@LAPTOP-LP9QU3RR: ~$ mkdir os_practical7
kruthi@LAPTOP-LP9QU3RR: ~$ cd os_practical7
kruthi@LAPTOP-LP9QU3RR: ~/os_practical7$ nano memory_layout.c
kruthi@LAPTOP-LP9QU3RR: ~/os_practical7$ ls
memory_layout.c
kruthi@LAPTOP-LP9QU3RR: ~/os_practical7$ gcc memory_layout.c -o m
emory_layout
kruthi@LAPTOP-LP9QU3RR: ~/os_practical7$ ls
memory_layout  memory_layout.c
kruthi@LAPTOP-LP9QU3RR: ~/os_practical7$ ./memory_layout
===== MEMORY ADDRESS SPACE =====
Process ID           : 732
Code address         : 0x5f7a037a2229
Global variable      : 0x5f7a037a5010
Static variable      : 0x5f7a037a5014
Heap variable        : 0x5f7a04b4f2a0
Stack variable       : 0x7ffc5b334fcc

===== VARIABLE VALUES =====
Global value         : 100
Static value         : 200
Heap value           : 400
Stack value          : 300

Process is running...
Press Enter to exit.

kruthi@LAPTOP-LP9QU3RR: ~/os_practical7$ ps -p 732
  PID TTY          TIME CMD
kruthi@LAPTOP-LP9QU3RR: ~/os_practical7$ |
```

```

kruthi@LAPTOP-LP9QU3RR: ~ × + ▾
kruthi@LAPTOP-LP9QU3RR:~$ cat /proc/845/maps
cat: /proc/845/maps: No such file or directory
kruthi@LAPTOP-LP9QU3RR:~$ ps -ef | grep memory_layout
kruthi      566      522  0 14:25 pts/2    00:00:00 grep --color
=auto memory_layout
kruthi@LAPTOP-LP9QU3RR:~$ cd ~/os_practical7
./memory_layout
===== MEMORY ADDRESS SPACE =====
Process ID      : 569
Code address    : 0x5ce6cb282229
Global variable : 0x5ce6cb285010
Static variable : 0x5ce6cb285014
Heap variable   : 0x5ce6d947f2a0
Stack variable  : 0x7ffeb2a0b6dc

===== VARIABLE VALUES =====
Global value    : 100
Static value    : 200
Heap value      : 400
Stack value     : 300

Process is running...
Press Enter to exit.
^C
kruthi@LAPTOP-LP9QU3RR:~/os_practical7$ ps -ef | grep '[m]emory_
layout'
kruthi      570      311  0 14:25 pts/0    00:00:00 ./memory_lay
out
kruthi@LAPTOP-LP9QU3RR:~/os_practical7$ ps -ef | grep '[m]emory_
layout'
kruthi      570      311  0 14:25 pts/0    00:00:00 ./memory_lay
out
kruthi@LAPTOP-LP9QU3RR:~/os_practical7$ ps -p 900
    PID TTY          TIME CMD
kruthi@LAPTOP-LP9QU3RR:~/os_practical7$ cat /proc/900/maps
cat: /proc/900/maps: No such file or directory
kruthi@LAPTOP-LP9QU3RR:~/os_practical7$ |

```



kruthi@LAPTOP-LP9QU3RR: ~



Process is running...
Press Enter to exit.

^C

kruthi@LAPTOP-LP9QU3RR:~/os_practical7\$ ps -ef | grep '[m]emory_layout'

kruthi 570 311 0 14:25 pts/0 00:00:00 ./memory_layout

kruthi@LAPTOP-LP9QU3RR:~/os_practical7\$ ps -ef | grep '[m]emory_layout'

kruthi 570 311 0 14:25 pts/0 00:00:00 ./memory_layout

kruthi@LAPTOP-LP9QU3RR:~/os_practical7\$ ps -p 900

PID TTY TIME CMD

kruthi@LAPTOP-LP9QU3RR:~/os_practical7\$ cat /proc/900/maps

cat: /proc/900/maps: No such file or directory

kruthi@LAPTOP-LP9QU3RR:~/os_practical7\$ ps -p 570

PID TTY TIME CMD

570 pts/0 00:00:00 memory_layout

kruthi@LAPTOP-LP9QU3RR:~/os_practical7\$ cat /proc/570/maps

63e779b16000-63e779b17000 r--p 00000000 08:30 59825

63e779b17000-63e779b18000 r-xp 00001000 08:30 59825

63e779b18000-63e779b19000 r--p 00002000 08:30 59825

63e779b19000-63e779b1a000 r--p 00002000 08:30 59825

63e779b1a000-63e779b1b000 rw-p 00003000 08:30 59825

63e7ace5c000-63e7ace7d000 rw-p 00000000 00:00 0

7b01de800000-7b01de828000 r--p 00000000 08:30 18227

7b01de828000-7b01de9b1000 r-xp 00028000 08:30 18227

7b01de9b1000-7b01dea00000 r--p 001b1000 08:30 18227

7b01dea00000-7b01dea04000 r--p 001ff000 08:30 18227

7b01dea04000-7b01dea06000 rw-p 00203000 08:30 18227

7b01dea06000-7b01dea13000 rw-p 00000000 00:00 0

7b01dea6f000-7b01dea72000 rw-p 00000000 00:00 0

7b01dea78000-7b01dea7a000 rw-p 00000000 00:00 0

7b01dea7a000-7b01dea7e000 r--p 00000000 00:00 0

7b01dea7e000-7b01dea80000 r--p 00000000 00:00 0

7b01dea80000-7b01dea82000 r-xp 00000000 00:00 0

7b01dea82000-7b01dea83000 r--p 00000000 08:30 18224

7b01dea83000-7b01deaae000 r-xp 00001000 08:30 18224

7b01deaae000-7b01deab8000 r--p 0002c000 08:30 18224

7b01deab8000-7b01deaba000 r--p 00036000 08:30 18224

7b01deaba000-7b01deabc000 rw-p 00038000 08:30 18224

7ffcd4209000-7ffcd422a000 rw-p 00000000 00:00 0

kruthi@LAPTOP-LP9QU3RR:~/os_practical7\$

/home/kruthi/os_practical7/memory_layout

/home/kruthi/os_practical7/memory_layout

/home/kruthi/os_practical7/memory_layout

/home/kruthi/os_practical7/memory_layout

/home/kruthi/os_practical7/memory_layout

[heap]

/usr/lib/x86_64-linux-gnu/libc.so.6

/usr/lib/x86_64-linux-gnu/libc.so.6

/usr/lib/x86_64-linux-gnu/libc.so.6

/usr/lib/x86_64-linux-gnu/libc.so.6

/usr/lib/x86_64-linux-gnu/libc.so.6

[vvar]

[vvar_vclock]

[vdso]

/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2

/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2

/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2

/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2

/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2

[stack]



kruthi@LAPTOP-LP9QU3RR: ~



```
ome/kruthi/os_practical7/memory_layout
63e779b18000-63e779b19000 r--p 00002000 08:30 59825
ome/kruthi/os_practical7/memory_layout
63e779b19000-63e779b1a000 r--p 00002000 08:30 59825
ome/kruthi/os_practical7/memory_layout
63e779b1a000-63e779b1b000 rw-p 00003000 08:30 59825
ome/kruthi/os_practical7/memory_layout
63e7ace5c000-63e7ace7d000 rw-p 00000000 00:00 0
eap]
7b01de800000-7b01de828000 r--p 00000000 08:30 18227
sr/lib/x86_64-linux-gnu/libc.so.6
7b01de828000-7b01de9b1000 r-xp 00028000 08:30 18227
sr/lib/x86_64-linux-gnu/libc.so.6
7b01de9b1000-7b01dea00000 r--p 001b1000 08:30 18227
sr/lib/x86_64-linux-gnu/libc.so.6
7b01dea00000-7b01dea04000 r--p 001ff000 08:30 18227
sr/lib/x86_64-linux-gnu/libc.so.6
7b01dea04000-7b01dea06000 rw-p 00203000 08:30 18227
sr/lib/x86_64-linux-gnu/libc.so.6
7b01dea06000-7b01dea13000 rw-p 00000000 00:00 0
7b01dea6f000-7b01dea72000 rw-p 00000000 00:00 0
7b01dea78000-7b01dea7a000 rw-p 00000000 00:00 0
7b01dea7a000-7b01dea7e000 r--p 00000000 00:00 0
var]
7b01dea7e000-7b01dea80000 r--p 00000000 00:00 0
var_vclock]
7b01dea80000-7b01dea82000 r-xp 00000000 00:00 0
dso]
7b01dea82000-7b01dea83000 r--p 00000000 08:30 18224
sr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7b01dea83000-7b01deaae000 r-xp 00001000 08:30 18224
sr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7b01deaae000-7b01deab8000 r--p 0002c000 08:30 18224
sr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7b01deab8000-7b01deaba000 r--p 00036000 08:30 18224
sr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7b01deaba000-7b01deabc000 rw-p 00038000 08:30 18224
sr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ffcd4209000-7ffcd422a000 rw-p 00000000 00:00 0
tack]
kruthi@LAPTOP-LP9QU3RR:~/os_practical7$ |
```



```
PID TTY          TIME CMD
kruthi@LAPTOP-LP9QU3RR:~/os_practical7$ cat /proc/900/maps
cat: /proc/900/maps: No such file or directory
kruthi@LAPTOP-LP9QU3RR:~/os_practical7$ ps -p 570
```

```
PID TTY          TIME CMD
570 pts/0      00:00:00 memory_layout
kruthi@LAPTOP-LP9QU3RR:~/os_practical7$ cat /proc/570/maps
```

```

63e779b16000-63e779b17000 r-rp 00000000 08:30 59825
63e779b17000-63e779b18000 r-xp 00001000 08:30 59825
63e779b18000-63e779b19000 r-rp 00002000 08:30 59825
63e779b19000-63e779b1a000 r-rp 00002000 08:30 59825
63e779b1a000-63e779b1b000 rw-p 00003000 08:30 59825
63e7ace5c000-63e7ace7d000 rw-p 00000000 00:00 0
7b01de800000-7b01de828000 r-rp 00000000 08:30 18227
7b01de828000-7b01de9b1000 r-xp 00023000 08:30 18227
7b01de9b1000-7b01dea00000 r-rp 001b1000 08:30 18227
7b01dea00000-7b01dea04000 r-rp 001f3000 08:30 18227
7b01dea04000-7b01dea06000 rw-p 00203000 08:30 18227
7b01dea06000-7b01dea13000 rw-p 00000000 00:00 0
7b01dea13000-7b01dea72000 rw-p 00000000 00:00 0
7b01dea72000-7b01dea7a000 rw-p 00000000 00:00 0
7b01dea7a000-7b01dea7e000 r-rp 00000000 00:00 0
7b01dea7e000-7b01dea80000 r-rp 00000000 00:00 0
7b01dea80000-7b01dea82000 r-xp 00000000 00:00 0
7b01dea82000-7b01dea83000 r-rp 00000000 08:30 18224
7b01dea83000-7b01dea8e000 r-xp 00001000 08:30 18224
7b01dea8e000-7b01dea8f000 r-rp 0002c000 08:30 18224
7b01dea8f000-7b01deaba000 rw-p 00036000 08:30 18224
7b01deaba000-7b01deabc000 rw-p 00038000 08:30 18224
7ffcfd4209000-7ffcfd422a000 rw-p 00000000 00:00 0

```

```
kruthi@LAPTOP-LP9QU3RR:~/os_practical7$ pmap 570
570:      ./memory_layout
000063e779b16000      4K r----- memory_layout
000063e779b17000      4K r-x-- memory_layout
000063e779b18000      4K r----- memory_layout
000063e779b19000      4K r----- memory_layout
000063e779b1a000      4K rw----- memory_layout
000063e7ace5c000     132K rw----- [ anon ]
00007b01de800000     160K r----- libc.so.6
00007b01de828000     1572K r-x-- libc.so.6
00007b01de9b1000     316K r----- libc.so.6
00007b01dea00000      16K r----- libc.so.6
00007b01dea04000       8K rw----- libc.so.6
00007b01dea06000       52K rw----- [ anon ]
00007b01dea6f000      12K rw----- [ anon ]
00007b01dea78000       8K rw----- [ anon ]
00007b01dea7a000      16K r----- [ anon ]
00007b01dea7e000       8K r----- [ anon ]
00007b01dea80000       8K r-x-- [ anon ]
00007b01dea82000       4K r----- ld-linux-x86-64.so.2
00007b01dea83000     172K r-x-- ld-linux-x86-64.so.2
00007b01deaae000     40K r----- ld-linux-x86-64.so.2
00007b01deab0000       8K r----- ld-linux-x86-64.so.2
00007b01deaba000       8K rw----- ld-linux-x86-64.so.2
00007ffcd4209000     132K rw----- [ stack ]
total                2692K
```

```
hruith@LAPTOP-PPQJ3RR:/~/os_practical$ cat /proc/570/maps
63e779b16000-63e779b17000 r-p 00000000 08:30 59825
63e779b17000-63e779b18000 r-xp 00001000 08:30 59825
63e779b18000-63e779b19000 r-p 00002000 08:30 59825
63e779b19000-63e779b1a000 r-p 00002000 08:30 59825
63e779b1a000-63e779b1b000 rw-p 00003000 08:30 59825
63e7ace5c000-63e7ace74000 rw-p 00000000 00:00 0
7b01de800000-7b01de828000 r-p 00000000 08:30 18227
7b01de828000-7b01de9b1000 r-xp 00028000 08:30 18227
7b01de9b1000-7b01dea00000 r-p 001b1000 08:30 18227
7b01dea00000-7b01dea04000 r-p 001ff000 08:30 18227
7b01dea04000-7b01dea06000 rw-p 00203000 08:30 18227
7b01dea06000-7b01dea13000 rw-p 00000000 00:00 0
7b01dea6f000-7b01dea72000 rw-p 00000000 00:00 0
7b01dea78000-7b01dea7a000 r-p 00000000 00:00 0
7b01dea7a000-7b01dea7e000 r-p 00000000 00:00 0
7b01dea7e000-7b01dea80000 r-p 00000000 00:00 0
7b01dea80000-7b01dea82000 r-xp 00000000 00:00 0
7b01dea82000-7b01dea83000 r-p 00000000 08:30 18224
7b01dea83000-7b01deaae000 r-xp 00001000 08:30 18224
7b01deaae000-7b01deab8000 r-p 0002c000 08:30 18224
7b01deab8000-7b01deaba000 r-p 00036000 08:30 18224
7b01deaba000-7b01deabc000 rw-p 00038000 08:30 18224
7ffc42409000-7ffc4242a000 r-p 00000000 00:00 0
```

```
kruthi@LAPTOP-LP9QU3RR:~/os_practical7$
```

```
/home/kruthi/os_practical7/memory_layout
/home/kruthi/os_practical7/memory_layout
/home/kruthi/os_practical7/memory_layout
/home/kruthi/os_practical7/memory_layout
/home/kruthi/os_practical7/memory_layout
[heap]
/usr/lib/x86_64-linux-gnu/libc.so.6
/usr/lib/x86_64-linux-gnu/libc.so.6
/usr/lib/x86_64-linux-gnu/libc.so.6
/usr/lib/x86_64-linux-gnu/libc.so.6
/usr/lib/x86_64-linux-gnu/libc.so.6
[vvar]
[vvar_vclock]
[ydso]
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
[stack]
```

```
[vvar]
[vvar_vclock]
[ydsos]
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
[stack]
```

```
/home/kruthi/os_practical7/memory_layout
/home/kruthi/os_practical7/memory_layout
/home/kruthi/os_practical7/memory_layout
/home/kruthi/os_practical7/memory_layout
/home/kruthi/os_practical7/memory_layout
[heap]
/usr/lib/x86_64-linux-gnu/libc.so.6
/usr/lib/x86_64-linux-gnu/libc.so.6
/usr/lib/x86_64-linux-gnu/libc.so.6
/usr/lib/x86_64-linux-gnu/libc.so.6
/usr/lib/x86_64-linux-gnu/libc.so.6
```

```
[vvar]
[vvar_vclock]
[vdsso]
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
[stack]
```